

Recursive Ontological Multiplicity Operator (Λ/Ξ) — Formalization

0. Purpose

This document formalizes the **Λ/Ξ evolution law** with **prime-resolved multiplicity** acting on a **carrier** that includes both **state** and **relations**, with **lawfulness** enforced by a **resonance functional**. The aim is (i) mathematical clarity, (ii) implementability, and (iii) a clean stability window.

1. Mathematical setting

1.1 Carrier (state + relations)

Let

$$\mathcal{M} := \mathcal{H}_x \times \mathcal{H}_H$$

where:

- $\mathcal{H}_x$ is a (real or complex) Banach/Hilbert space of **state** variables x .
- $\mathcal{H}_H$ is a Banach space of **relational encodings** H (e.g., weighted adjacency matrices, incidence tensors, simplicial/hypergraph weights, relation embeddings). [9]

Define the product metric / norm

$$\|(x, H)\|_{\mathcal{M}} := \|x\|_x + \gamma \|H\|, \quad \gamma > 0.$$

Write $\mathcal{M}_t := (x_t, H_t)$.

1.2 Transform operator

Let

$$T: \mathcal{M} \rightarrow \mathcal{M}, \quad T(\mathcal{M}) = (T_x(\mathcal{M}), T_H(\mathcal{M}))$$

be **globally Lipschitz** with constant L_T :

$$\|T(\mathcal{M}) - T(\mathcal{M}')\|_{\mathcal{M}} \leq L_T \|\mathcal{M} - \mathcal{M}'\|_{\mathcal{M}}.$$

2. Prime-resolved structure

2.1 Prime indexing

Let $\mathbb{P}$ denote the set of primes. All sums indexed by $p \in \mathbb{P}$ are assumed absolutely convergent where required.

2.2 Prime-sector gates (projectors)

To ensure sector separation, define prime gates from a chosen commuting normal operator A (or commuting family) on $\mathcal{H}_x$.

Choose a measurable partition $\{\Omega_p\}_{p \in \mathbb{P}}$ of $\sigma(A)$ and define

$$P_p := \chi_{\Omega_p}(A).$$

This is an application of spectral/functional calculus for commuting normal operators [3,4].

Then

$$P_p P_q = 0 \quad (p \neq q), \quad \sum_p P_p = I, \quad [P_p, P_q] = 0, \quad [P_p, A] = 0.$$

Similarly, define gates Q_p acting on $\mathcal{H}_H$ (often by choosing an operator B on relation space and setting $Q_p = \chi_{\Theta_p}(B)$).

3. Dynamics: the Λ/Ξ evolution law

3.1 Prime-decomposed evolution operator (Ξ)

Define the state evolution operator

$$\Xi(t) := \sum_{p \in \mathbb{P}} w_p(t) U_p(t)$$

acting on $\mathcal{H}_x$, where $U_p(t) \in \mathcal{B}(\mathcal{H}_x)$ and $w_p(t) \in \mathbb{C}$ (or $\mathbb{R}$).

Assume boundedness, e.g.

$$\sup_t \sum_p |w_p(t)| \|U_p(t)\| < \infty$$

so $\Xi(t)$ is well-defined and bounded.

3.2 Prime-resolved multiplicity operator (Λ)

Define the multiplicity feedback operator on the carrier by

$$\Lambda(t)T(\mathcal{M}) := \sum_{p \in \mathbb{P}} \lambda_p(t) \bigl(P_p T_x(\mathcal{M}), Q_p T_H(\mathcal{M}) \bigr),$$

where $\lambda_p(t) \in \mathbb{R}$ (or complex, if permitted by the model).

Special case (global multiplicity knob): $\lambda_p(t) \equiv \Lambda_m(t)$ and $P_p = I, Q_p = I$.

3.3 Discrete-time carrier update

Define the one-step map $\Phi_t : \mathcal{M} \rightarrow \mathcal{M}$ by

$$\Phi_t(x, H) := \bigl(\Xi(t)x, H \bigr) + \Lambda(t)T(x, H).$$

Then the $\Lambda \Xi$ evolution is

$$\mathcal{M}_{t+1} = \Phi_t(\mathcal{M}_t).$$

4. Stability window (contraction budget)

4.1 Operator-norm gain bound

Define the effective feedback gain bound

$$G(t) := L_T \sum_p |\lambda_p(t)| \max\{|P_p|, |Q_p|\}.$$

Assume $\|P_p\|, \|Q_p\| \leq 1$ when they are orthogonal projections.

4.2 Uniform contraction condition

Assume

$$\sup_t |\Xi(t)| \leq 1 - \epsilon \quad (\epsilon > 0)$$

and enforce the **gain budget**

$$\sup_t G(t) \leq \epsilon.$$

Then for all t , the map Φ_t is a contraction on $(\mathcal{M}, \|\cdot\|_{\mathcal{M}})$ with contraction ratio at most

$$\rho := \sup_t \bigl(|\Xi(t)| + G(t) \bigr) \leq 1 - \epsilon + \epsilon = 1.$$

To ensure strict contraction, strengthen to $\sup_t G(t) \leq \epsilon - \delta$ for some $\delta > 0$, yielding $\rho \leq 1 - \delta$.

5. Drift monitoring and safe self-correction

5.1 Drift observables

Let $\Phi(p, t)$ be a prime-sector observable (e.g., a phase, alignment, or coherence statistic derived from P_p -filtered state/relations). Define per-prime drift signals $\delta_p(t)$ and a global drift score $\delta_{\text{drift}}(t)$.

5.2 Projected gain update (budget-preserving)

Choose a step size $\eta > 0$ and a feedback function g that reduces gain when drift is large (e.g., g decreasing in δ_p). Update

$$\lambda_p(t+1) = \Pi_{[0, \lambda_p^{\max}]}(\lambda_p(t) + \eta g(\delta_p(t)))$$

where Π clips to an interval and $\lambda_p^{\max}$ is chosen so the gain budget holds:

$$\sum_p \lambda_p^{\max} \max\{|P_p|, |Q_p|\} \leq \epsilon \delta$$

This ensures self-correction **cannot** violate contraction. Related projected/proximal update schemes are standard; see [7,8].

6. Resonance validation (lawfulness criterion)

6.1 Resonance functional

Define a bounded functional

$$R: \mathcal{M} \rightarrow [0, 1]$$

interpreted as a measure of coherence / agreement with constraints or data.

Implementation-friendly canonical form:

$$R(\mathcal{M}) := \exp\left(-\beta \|\mathcal{L}(\mathcal{M})\| + \alpha \|\mathcal{L}\| + \kappa \|\mathcal{C}(\mathcal{M})\|\right)$$

with $\beta, \alpha, \kappa \geq 0$.

6.2 Resonant selection over operator words / edits

When $\Xi(t)$ is realized as an *operator word* (noncommuting compositions) and T_H proposes relational edits:

8. Minimal validation protocol

8.1 State-only experiment (no relations)

- Take $\mathcal{H}_x = \mathbb{C}^n$ with small n .
- Construct commuting P_p from diagonal A .
- Pick bounded contractions $U_p(t)$ and weights $w_p(t)$.
- Choose Lipschitz T_x (e.g., $\tanh$ applied componentwise plus a bounded linear map).
- Verify empirical contraction ratio and drift reduction under projected λ_p -updates.

8.2 Ontological (relations) upgrade

- Let H be a weighted hypergraph relaxation.
- Define T_H as a proximal/gradient step on a loss used inside resonance.
- Optionally round H after each epoch (outside contraction proof).
- Run resonance-based candidate selection and measure stability + interpretability of emergent motifs.

8.3 Cross-domain same-equation demo

Run the same Λ/Ξ core with the same gain budget logic on:

- denoising (signals),
- a simple recurrent cognitive task (belief + relations),
- complex amplitude toy dynamics, and compare when/why trajectories remain stable and coherent.

9. References

- [1] Ryan Van Gelder. *Am And $\Xi(t)$ — Working Specification*. Draft manuscript/PDF, 2025.
- [2] Ryan Van Gelder. *Multiplicity Operator Calculus*. Draft manuscript/PDF, November 5, 2025.
- [3] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*. Academic Press, 1972.
- [4] John B. Conway. *A Course in Functional Analysis*. 2nd ed., Springer, 1990.
- [5] Stefan Banach. “Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales.” *Fundamenta Mathematicae* 3(1): 133–181, 1922.
- [6] F. J. Dyson. “The S Matrix in Quantum Electrodynamics.” *Physical Review* 75(11): 1736–1755, 1949. DOI: 10.1103/PhysRev.75.1736.
- [7] Neal Parikh and Stephen Boyd. “Proximal Algorithms.” *Foundations and Trends in Optimization* 1(3): 127–239, 2014. DOI: 10.1561/2400000003.

[8] Heinz H. Bauschke and Patrick L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. Springer, 2011.

[9] Claude Berge. *Hypergraphs: Combinatorics of Finite Sets*. North-Holland Mathematical Library 45, North-Holland, 1989.

[10] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1966.

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